

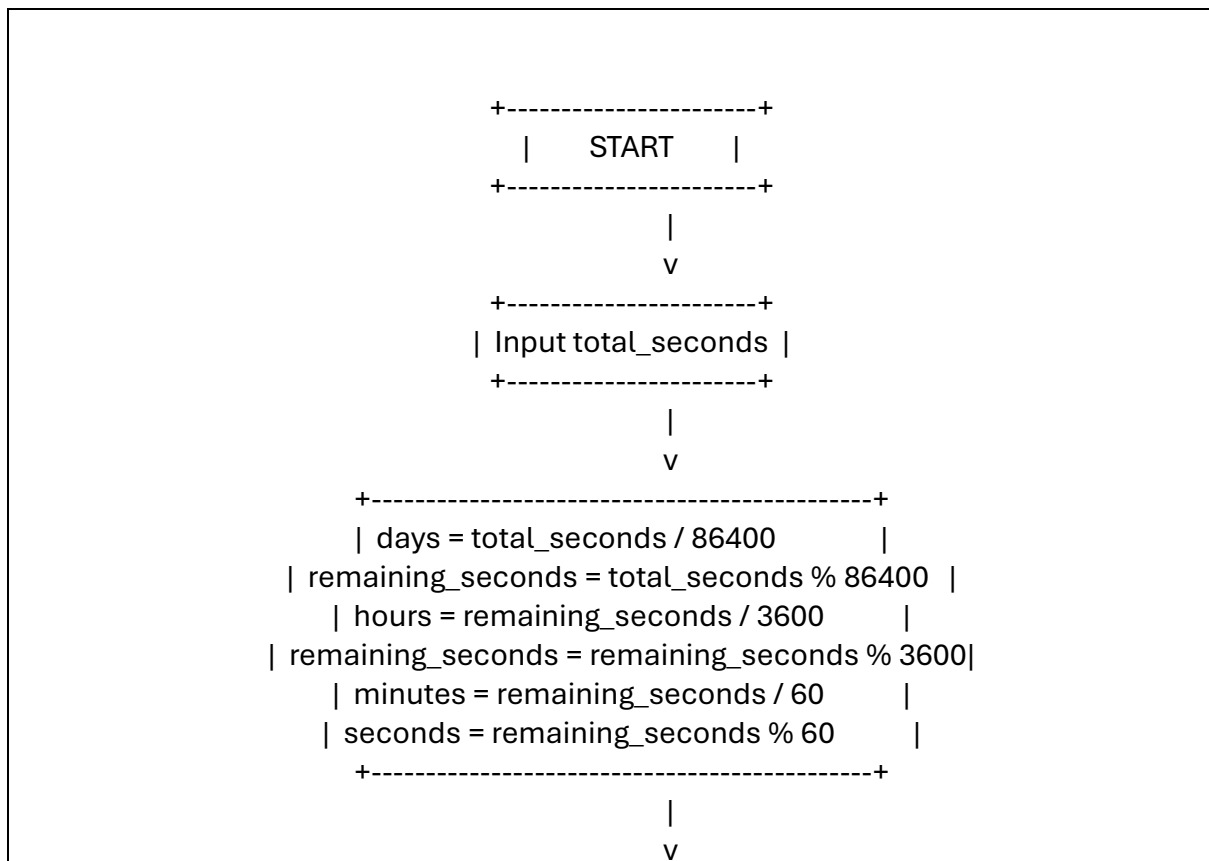
HDOC Day-10 Activity

Question 1: Seconds to Days, Hours, Minutes, and Seconds Converter

1. Problem Statement & Identification

- **Problem:** Input total time in seconds and convert it into days, hours, minutes, and remaining seconds.
- **Input:** Time in seconds (total_seconds).
- **Output:** Time represented in Days, Hours, Minutes, and Seconds.

2. Flowchart



```

+-----+
| Print days, hours, |
| minutes, seconds  |
+-----+
      |
      v
+-----+
|      END      |
+-----+

```

3.Pseudocode-

```

BEGIN
  READ total_seconds
  days <- total_seconds / 86400
  rem_sec <- total_seconds MOD 86400
  hours <- rem_sec / 3600
  rem_sec <- rem_sec MOD 3600
  minutes <- rem_sec / 60
  seconds <- rem_sec MOD 60
  PRINT days, hours, minutes, seconds
END

```

4. Algorithm

1. Start
2. Input total_seconds.
3. Compute days = total_seconds / 86400 and update remaining_seconds = total_seconds % 86400.
4. Compute hours = remaining_seconds / 3600 and update remaining_seconds = remaining_seconds % 3600.
5. Compute minutes = remaining_seconds / 60.
6. Compute seconds = remaining_seconds % 60.

7. Output days, hours, minutes, and seconds.

8. End

5. Test Case Table

Test Case	Input (total_seconds)	Expected Output	Actual Output	Status
1	90061	1 Days, 1 Hours, 1 Minutes, 1 Seconds	1 Days, 1 Hours, 1 Minutes , 1 Seconds	Pass
2	3661	0 Days, 1 Hours, 1 Minutes, 1 Seconds	0 Days, 1 Hours, 1 Minutes , 1 Seconds	Pass

Test Case	Input (total_seconds)	Expected Output	Actual Output	Status
3	86400	1 Days, 0 Hours, 0 Minutes, 0 Seconds	1 Days, 0 Hours, 0 Minutes, 0 Seconds	Pass

6. C Source Code

```
#include <stdio.h>

int main(void) {
    long total_seconds, remaining_seconds;
    int days, hours, minutes, seconds;

    printf("Enter time in seconds: ");
    if (scanf("%ld", &total_seconds) != 1 || total_seconds < 0) {
        printf("Invalid input!\n");
        return 1;
    }

    days = total_seconds / 86400;
    remaining_seconds = total_seconds % 86400;

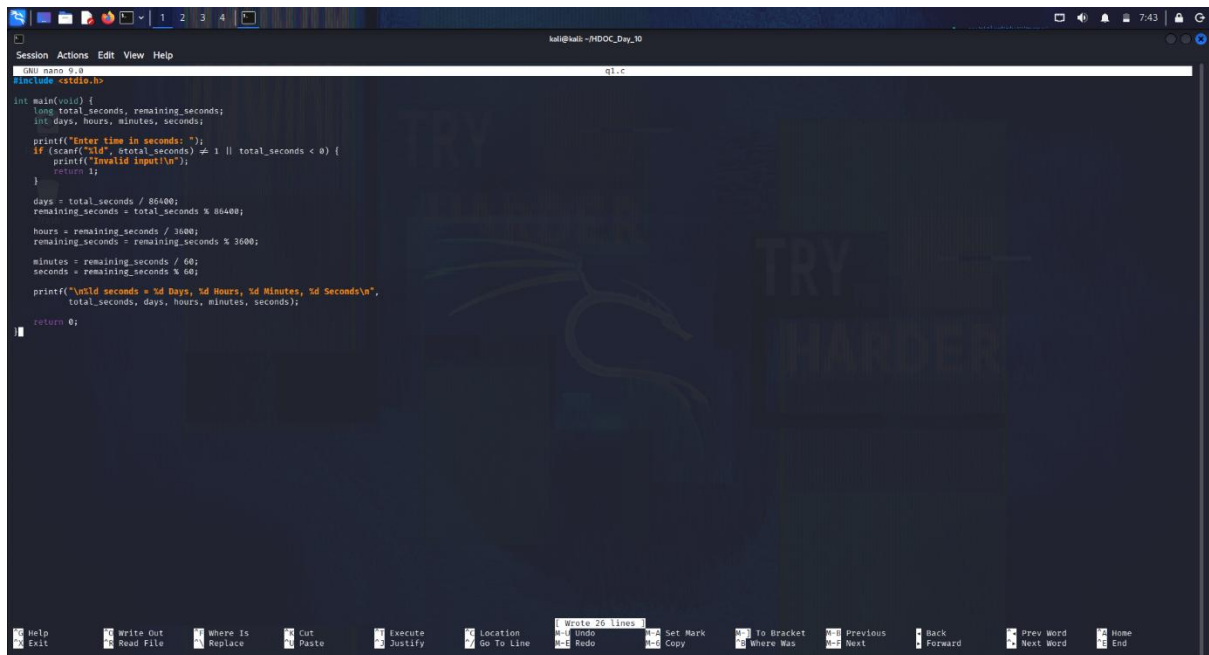
    hours = remaining_seconds / 3600;
    remaining_seconds = remaining_seconds % 3600;

    minutes = remaining_seconds / 60;
    seconds = remaining_seconds % 60;

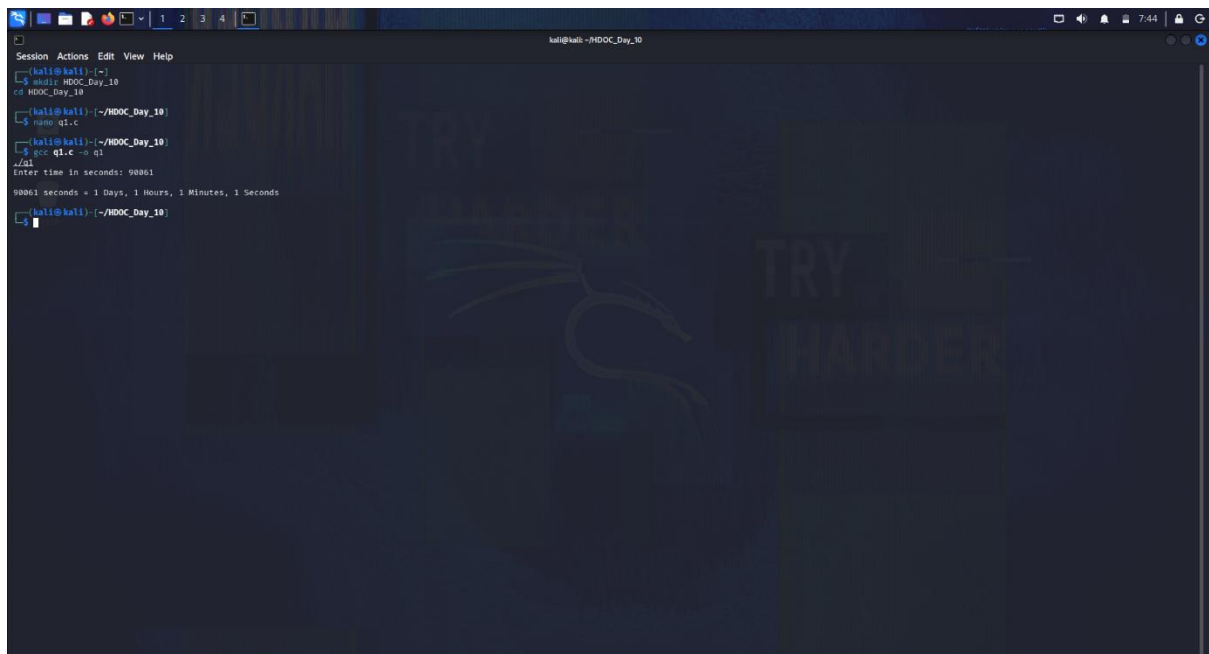
    printf("\n%ld seconds = %d Days, %d Hours, %d Minutes, %d Seconds\n",
```

```
total_seconds, days, hours, minutes, seconds);  
  
return 0;  
}
```

7. Compilation, Execution & Screenshots



```
Session Actions Edit View Help  
q1.c  
#include <stdio.h>  
  
int main(void) {  
    long total_seconds, remaining_seconds;  
    int days, hours, minutes, seconds;  
  
    printf("Enter time in seconds: ");  
    if (scanf("%ld", &total_seconds) != 1 || total_seconds < 0) {  
        printf("Invalid input!\n");  
        return 1;  
    }  
  
    days = total_seconds / 86400;  
    remaining_seconds = total_seconds % 86400;  
    hours = remaining_seconds / 3600;  
    remaining_seconds = remaining_seconds % 3600;  
    minutes = remaining_seconds / 60;  
    seconds = remaining_seconds % 60;  
  
    printf("Total seconds = %ld Days, %ld Hours, %ld Minutes, %ld Seconds\n",  
        total_seconds, days, hours, minutes, seconds);  
  
    return 0;  
}
```



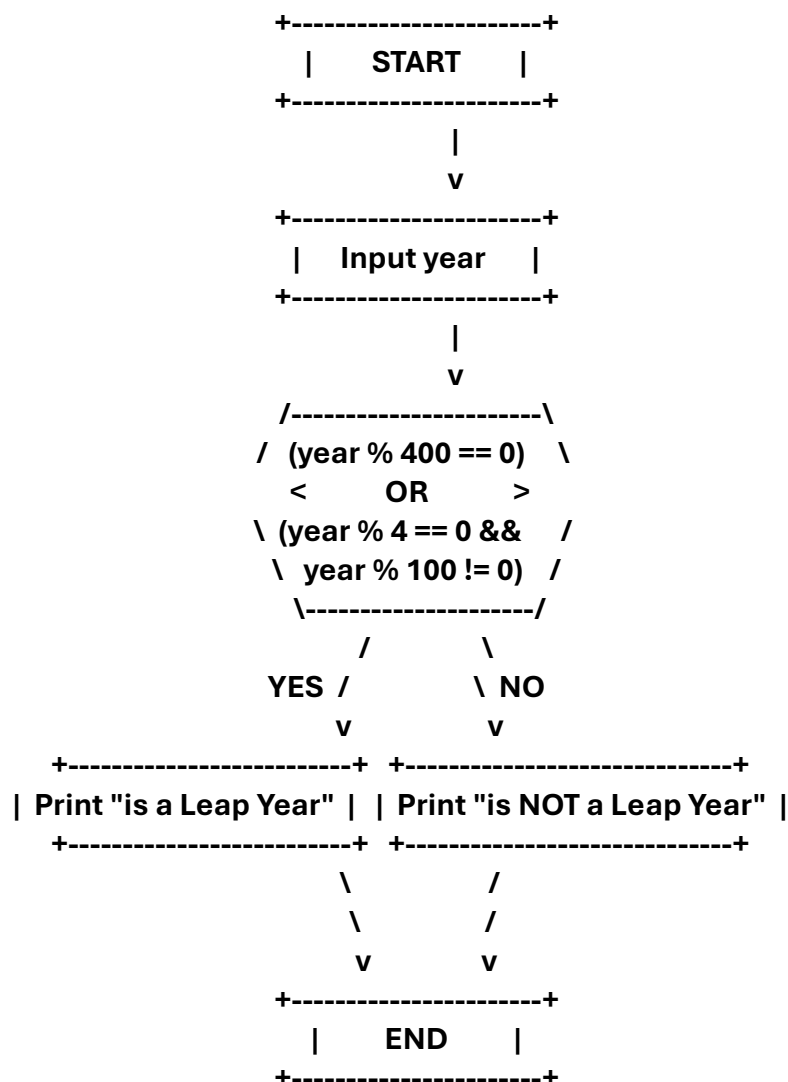
```
Session Actions Edit View Help  
[kali@kali:~]  
$ mkdir HDOOC_Day_10  
$ cd HDOOC_Day_10  
$ nano q1.c  
[kali@kali:~/HDOOC_Day_10]  
$ gcc q1.c -o q1  
./q1  
Enter time in seconds: 98861  
98861 seconds = 1 Days, 1 Hours, 1 Minutes, 1 Seconds  
[kali@kali:~/HDOOC_Day_10]
```

Question 2: Leap Year Checker

1. Problem Statement & Identification

- **Problem:** Input a year and determine whether it is a leap year.
- **Input:** Year (year).
- **Output:** Statement indicating whether the input year is a Leap Year or Not.

2. Flowchart



3. Pseudocode

```
BEGIN
  READ year
  IF (year MOD 400 == 0) OR (year MOD 4 == 0 AND year MOD 100 != 0) THEN
    PRINT "Leap Year"
  ELSE
    PRINT "NOT a Leap Year"
  END IF
END
```

4. Algorithm

1. Start

2. Input year.

3. Check if (year % 400 == 0) OR (year % 4 == 0 AND year % 100 != 0):

- **If True: Print "year is a Leap Year".**
- **If False: Print "year is NOT a Leap Year".**

4. End

5. Test Case Table

Test Case	Input (year)	Logic Evaluation	Expected Output	Actual Output	Status
1	2024	Divisible by 4, not by 100	2024 is a Leap Year	2024 is a Leap Year	Pass
2	1900	Divisible by 100, not by 400	1900 is NOT a Leap Year	1900 is NOT a Leap Year	Pass
3	2000	Divisible by 400	2000 is a Leap Year	2000 is a Leap Year	Pass

6. C Source Code

```
#include <stdio.h>

int main(void) {
    int year;

    printf("Enter a year: ");
    if (scanf("%d", &year) != 1 || year <= 0) {
        printf("Invalid input!\n");
        return 1;
    }
}
```



```

}

if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
    printf("%d is a Leap Year.\n", year);
} else {
    printf("%d is NOT a Leap Year.\n", year);
}

return 0;

```

7. Compilation, Execution & Screenshots

The screenshot shows a terminal window with the following content:

```

GNU nano 9.0
#include <stdio.h>

int main(void) {
    int year;

    printf("Enter a year: ");
    if (scanf("%d", &year) != 1 || year <= 0) {
        printf("Invalid input!\n");
        return 1;
    }

    if ((year % 400 == 0) || (year % 4 == 0 && year % 100 != 0)) {
        printf("1d is a Leap Year.\n", year);
    } else {
        printf("2d is NOT a Leap Year.\n", year);
    }

    return 0;
}

```

The terminal output shows the program execution with the input 2:

```

2
1d is a Leap Year.

```

The terminal window has a title bar that reads "kali@kali: ~/RDOC_Day_10". The bottom status bar shows various editor commands and the current line/line count (1/1).



```
kali@kali: ~/HDOOC_Day_10
Session Actions Edit View Help
kali@kali:~$ cd
kali@kali:~$ mkdir HDOOC_Day_10
kali@kali:~$ cd HDOOC_Day_10
kali@kali:~/HDOOC_Day_10$ nano q1.c
kali@kali:~/HDOOC_Day_10$ gcc q1.c -o q1
./q1
Enter time in seconds: 90001
90001 seconds = 1 Days, 1 Hours, 1 Minutes, 1 Seconds
kali@kali:~/HDOOC_Day_10$ nano q2.c
kali@kali:~/HDOOC_Day_10$ gcc q2.c -o q2
./q2
Enter a year: 2024
2024 is a Leap Year.
kali@kali:~/HDOOC_Day_10$
```